Strongly Correlated Ultracold gases

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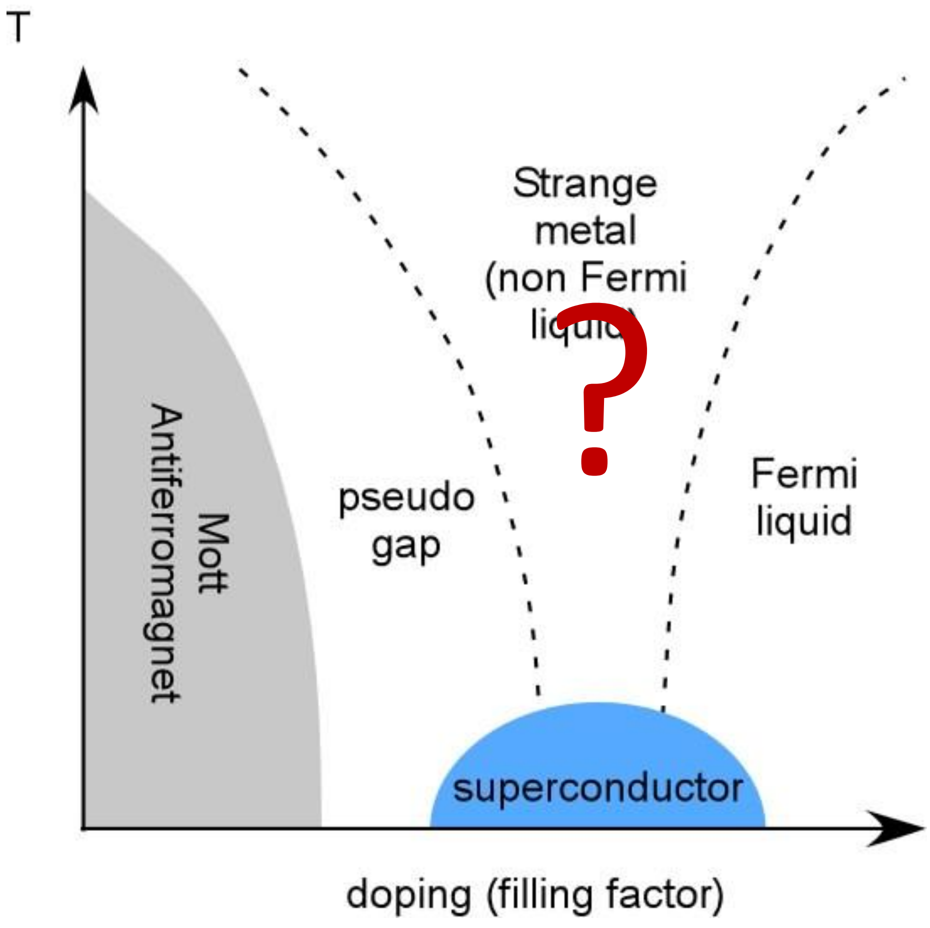
The Quantum Many-Body Problem

What is the gound state/equation of state of a given **quantum**-**many-body** hamiltonian H?

- Nuclear physics (nuclear shells;quark-gluon plasma)
- Astrophysics (neutron stars)
- Solid state physics

High-T_c superconductors

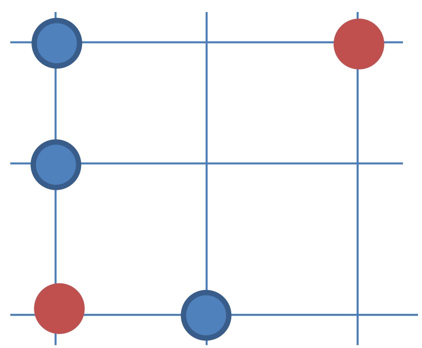
- Discovered by Bednorz and Mueller in 1987

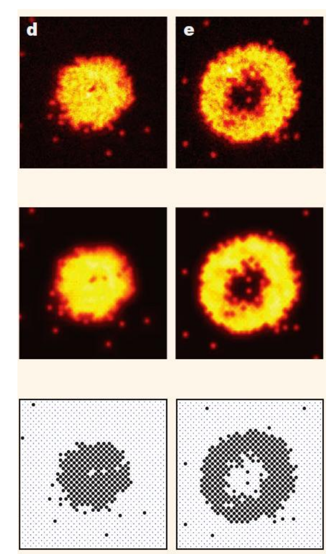


15 competing models!

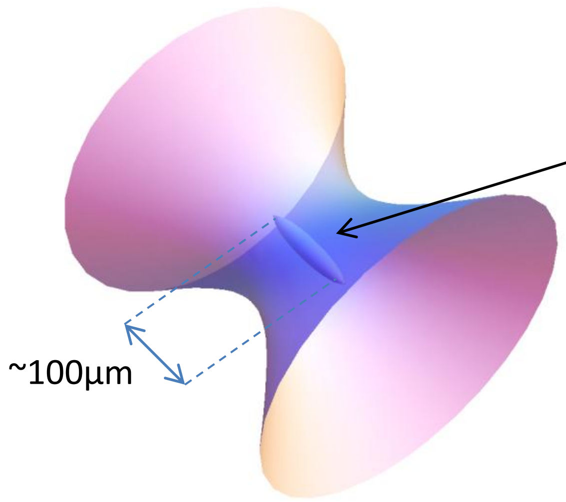
Quantum simulation using ultracold atoms

- **Goal:** test experimentally and quantitatively the predictions of quantum-many-body calculations on ultracold atom vapors.





Some numbers



$\sim 10^5$ atoms in an optical/magnetic harmonic trap

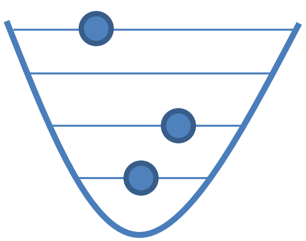
	Electrons in metal	Ultra cold atoms
Density	$10^{30}/m^3$	$10^{20}/m^3$
Mass	10^{-30}kg	10^{-26}kg
Fermi temperature$\sim n^{2/3}/m$	10^5K	1µK
T/T_F	$<10^{-5}$	$\sim 10^{-2}$

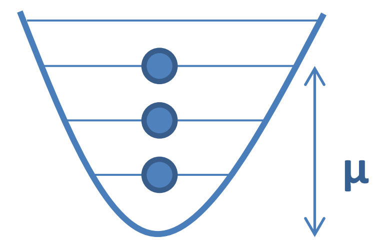
Trapped gases :
The ideal Fermi gas

The Fermi-Dirac distribution

Consider an ensemble of N spin polarized fermions at temperature T.

Grand canonical occupation number $$f(E) = \frac{1}{1+e^{\beta(E-\mu)}}$$





$\beta\mu$

0

$\beta\mu << -1$:
Boltzmann distribution

$$f(E) \simeq e^{\beta\mu} e^{-\beta E}$$

$\beta\mu >> 1$:
Zero-temperature Fermi distribution

$$f(E) \simeq \theta(\mu - E)$$

$\mu = E_F$=Fermi energy

The Zero-Temperature spin polarized Fermi gas

In a box of volume V density n=N/V: $D(E) = \frac{V}{2\pi^2}\sqrt{\frac{2m^3 E}{\hbar^6}}$

$$N = \int f(E) D(E) dE \Rightarrow E_F = \frac{\hbar^2 k_F^2}{2m} \quad \text{with} \quad k_F = (6\pi^2 n)^{1/3}$$

Rule of thumb: $\lambda_F = k_F^{-1}$ ~interparticle distance.

In a harmonic trap: $D(E) = \frac{E^2}{2(\hbar\bar{\omega})^3}$ with $\bar{\omega}^3 = \omega_x \omega_y \omega_z$

$$N = \int f(E) D(E) dE \Rightarrow E_F = \hbar\bar{\omega}(6N)^{1/3}$$

Energy of a trapped Fermi gas

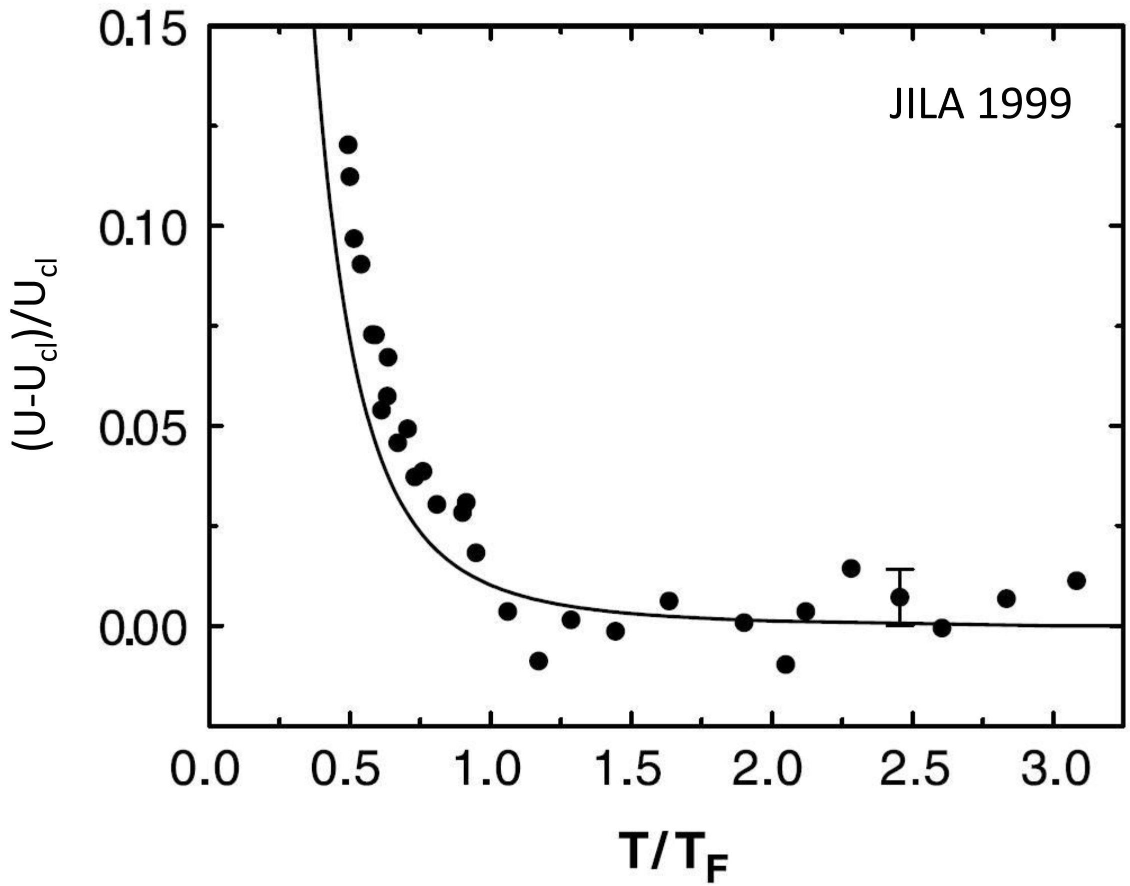
$$U = \int E f(E) D(E) dE$$

$T \gg T_F$: ideal Boltzmann gas (Equipartition Theorem)

$$U = U_{\text{clas}} = 3Nk_B T$$

$T \ll T_F$:

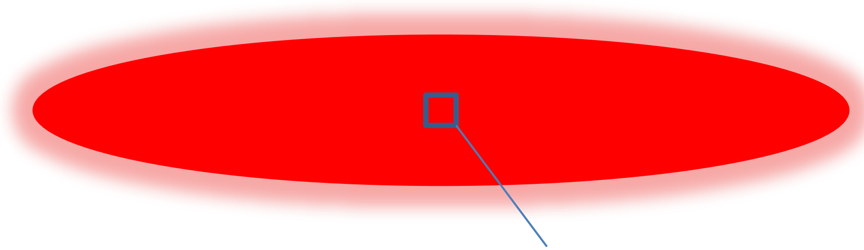
$$U = \frac{3}{4} N E_F$$



U measured by time of flight expansion and using virial theorem $U=2E_{kin}$. (F. Werner, PRA 2008)

Local Density Approximation

- What is the density profile of the cloud?

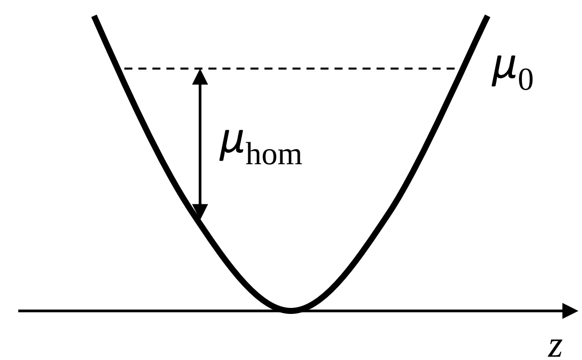


Mesoscopic volume where thermodynamical equilibrium is well defined (T(**r**),μ(**r**),P(**r**))

Equilibrium Condition: T,μ uniform.

Local Density Approximation:

$$\mu(\mathbf{r}) = \mu_{\mathrm{hom}}(n(\mathbf{r}),T) + V(\mathbf{r}) \equiv \mu_0$$



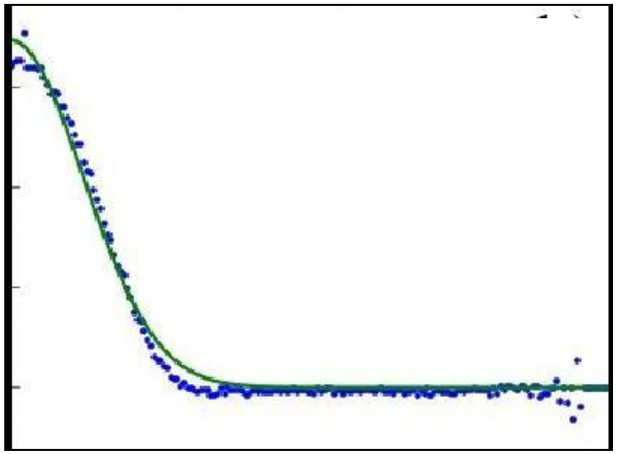
Note: this is the hydrostatic condition $\nabla P + n\nabla V = 0$ with the Gibbs-Duhem relation dP=ndμ at constant T.

Density profile of a trapped Fermi gas

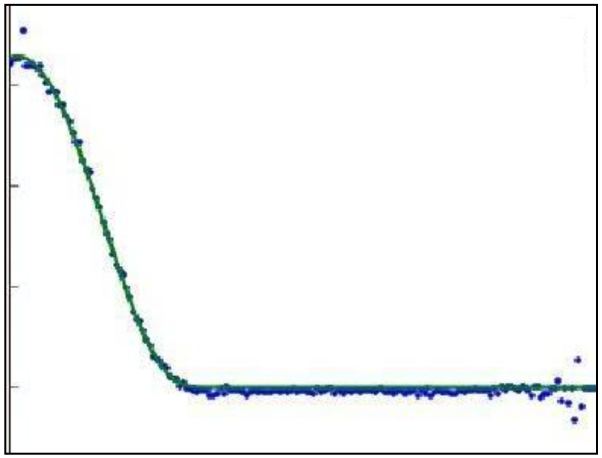
Zero temperature Fermi gas

$$\mu_{\mathrm{hom}}(n,T=0)=\frac{\hbar^2}{2m}(6\pi^2 n)^{2/3} \Rightarrow n(r)=\frac{1}{6\pi^2}\left[\frac{2m}{\hbar^2}\left(\mu_0-V(r)\right)\right]^{3/2}$$

Thomas-Fermi profile



Gaussian Fit



Thomas-Fermi Fit

N.B. $N=\int n(\mathbf{r})d^3\mathbf{r} \Rightarrow \mu_0=\hbar\bar{\omega}(6N)^{1/3}$, as before

Trapped gases :
The ideal Bose gas

The Bose-Einstein Condensation

Consider an ensemble of N spin polarized bosons at temperature T.

$$f(E) = \frac{1}{e^{\beta(E-\mu)} - 1}$$

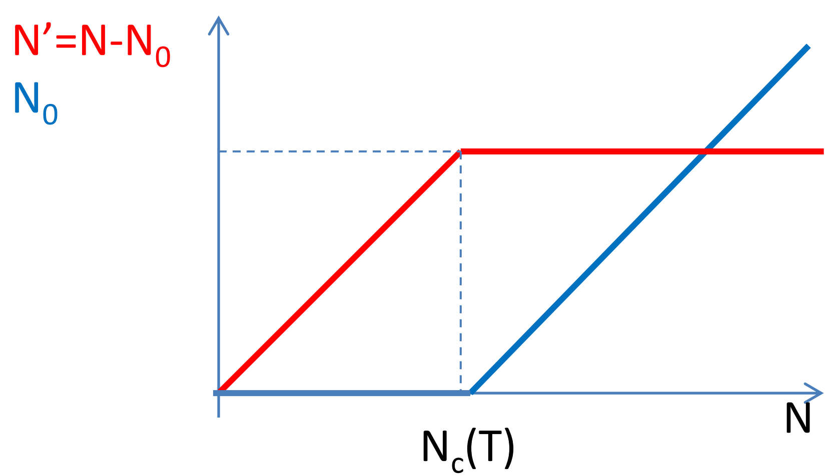
For a positive energy spectrum $f(E) > 0 \Rightarrow \mu < 0$

$$f(E) \leq \frac{1}{e^{\beta E} - 1} \quad \Rightarrow N \leq \int \frac{D(E)}{e^{\beta E} - 1} dE = N_c(T)$$

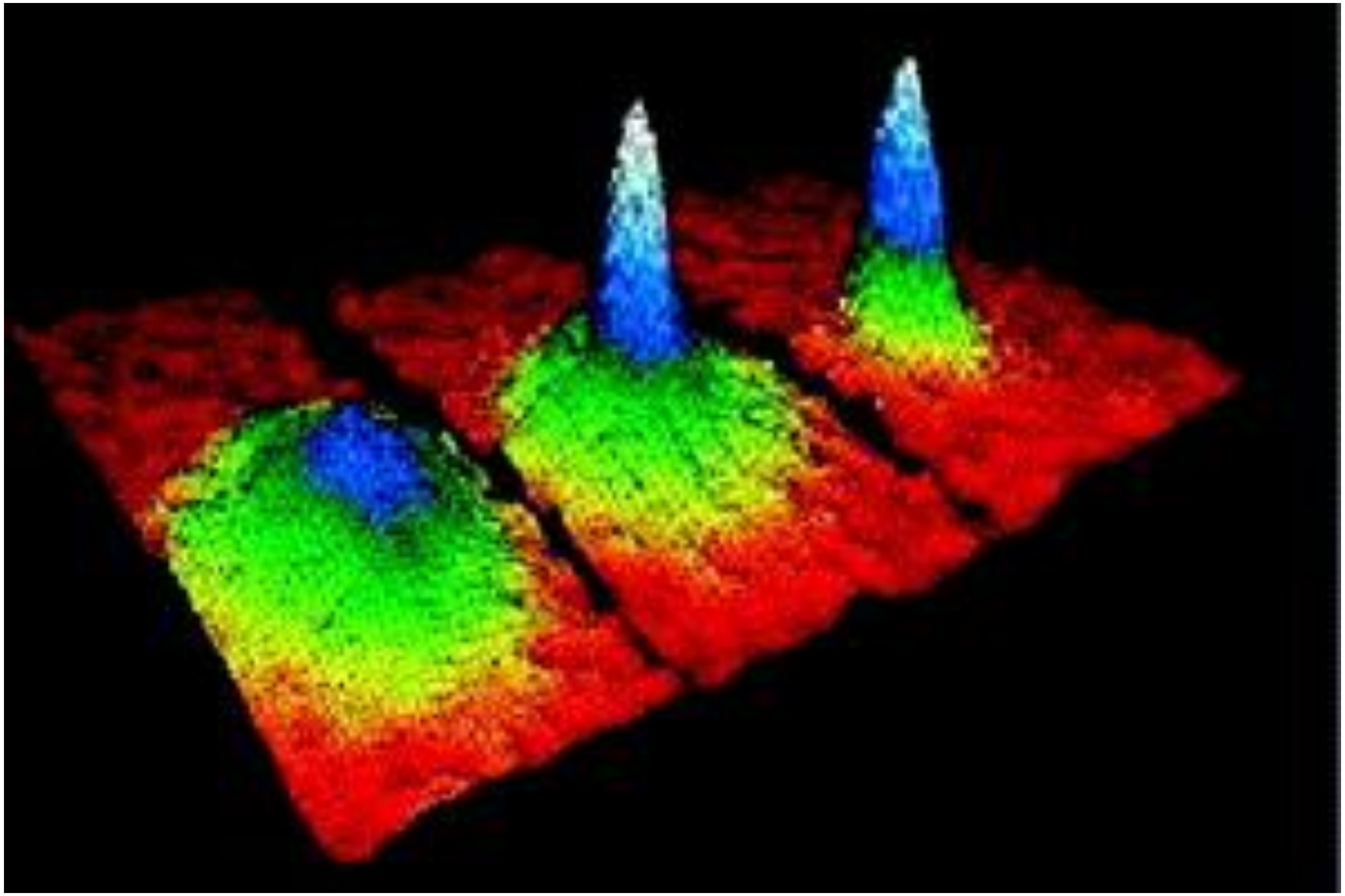
$$\left.\begin{array}{c} D(E) = CE^{\alpha} \\ \alpha > 0 \end{array}\right\} \Rightarrow N_c(T) \text{ is } \textbf{finite}$$

The system can only accomodate a finite number of particles. Above N_c, the bosons accumulate in their ground state: **Bose-Einstein Condensation**.

Filling at fixed temperature



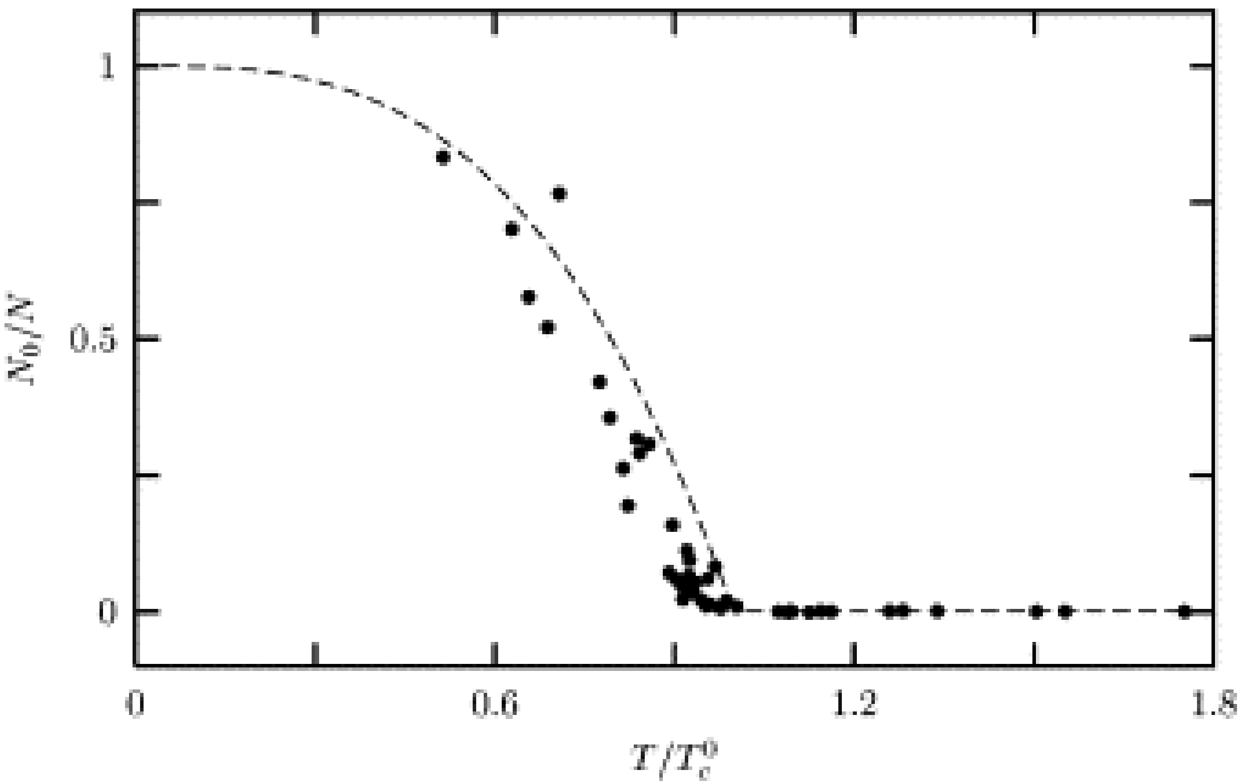
Bose-Einstein Condensation in Ultracold gases



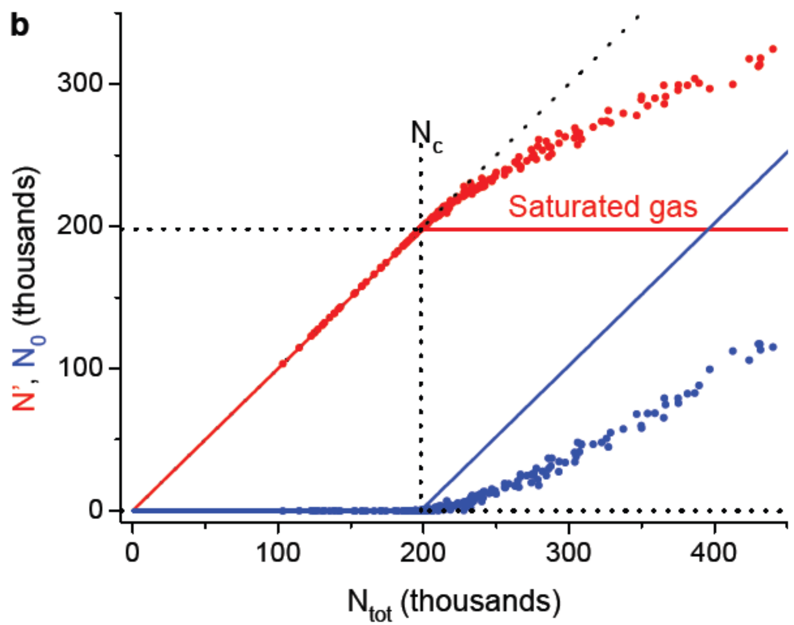
MIT, JILA 1995

Condensed fraction

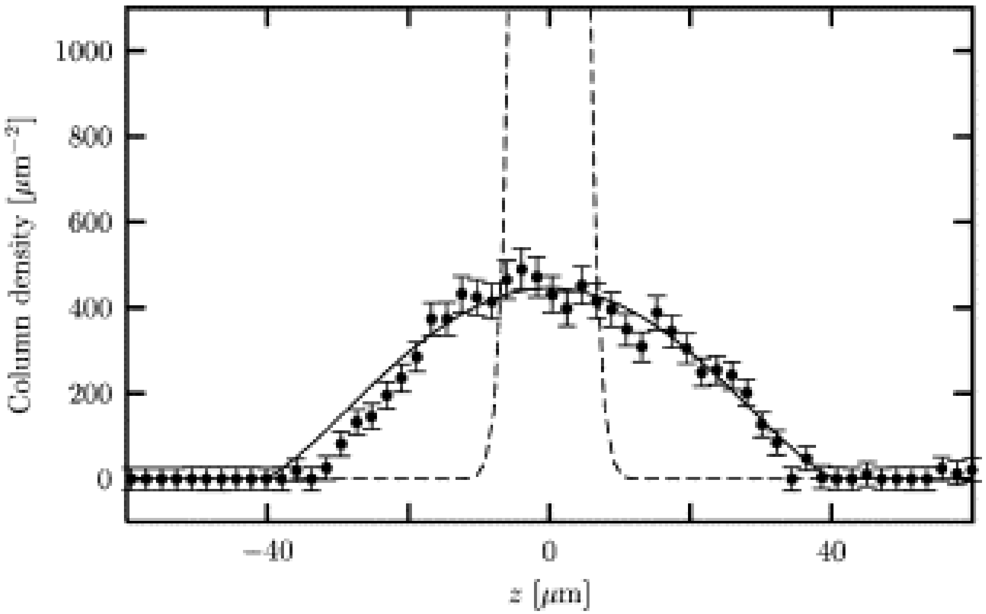
$$N_0 = N - N_c \Rightarrow \frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{\alpha+1}$$



Is a dilute BEC really an ideal gas?



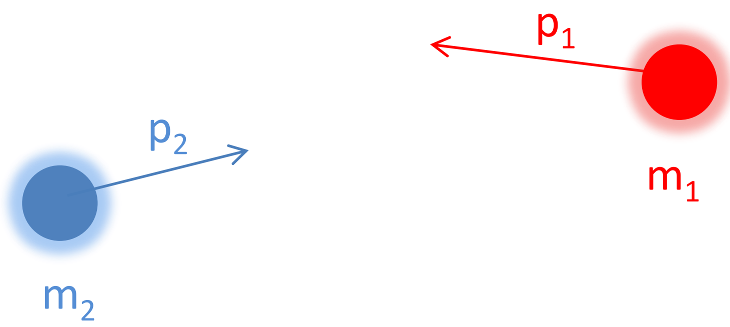
Phys. Rev. Lett. **106**, 230401 (2011)

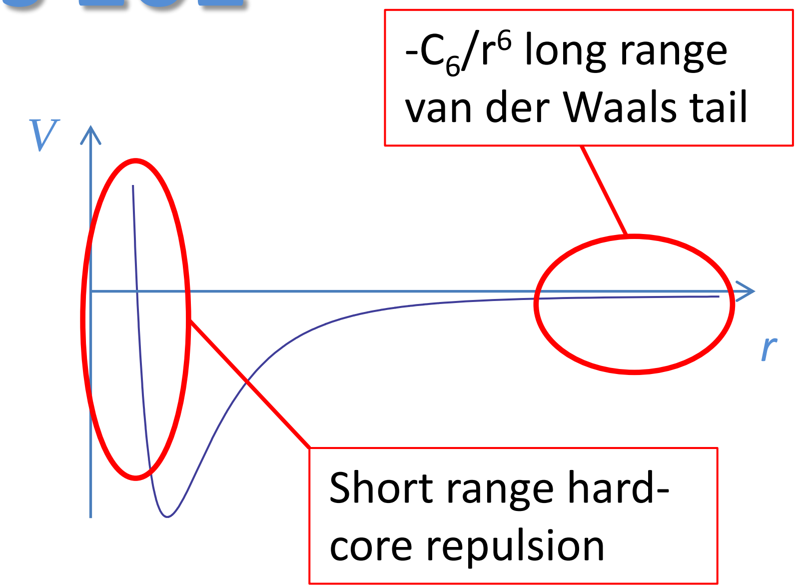


Rev. Mod. Phys. **71**, 463 (1999)

Ultracold collisions

Collisions 101





In the lab frame: $\hat{H} = \frac{\hat{p}_1^2}{2m_1} + \frac{\hat{p}_2^2}{2m_2} + V(|\hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2|)$

In the center of mass frame: $\hat{H} = \frac{\hat{p}^2}{2\mu} + V(|\hat{\mathbf{r}}|)$

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$$

$$\mathbf{p} = (\mathbf{p}_2 - \mathbf{p}_1)/2$$

$$\mu^{-1} = m_1^{-1} + m_2^{-1}$$

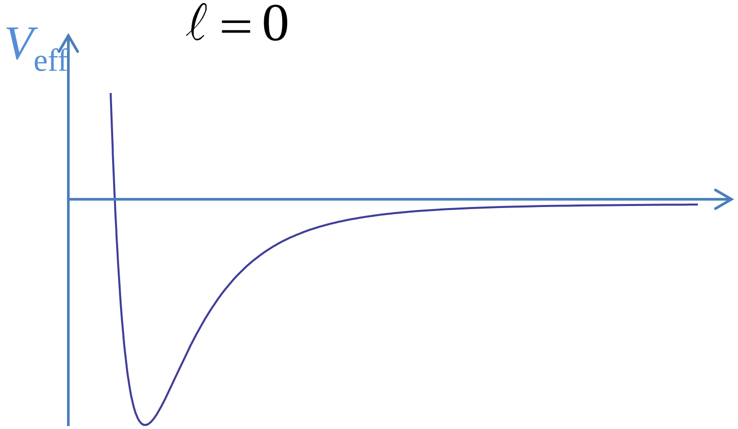
Stationary scattering states

Look for eigenstates of H $\psi(r,\theta,\varphi) = \dfrac{u(r)}{r} Y_\ell^m(\theta,\varphi)$ with $u(0) = 0$

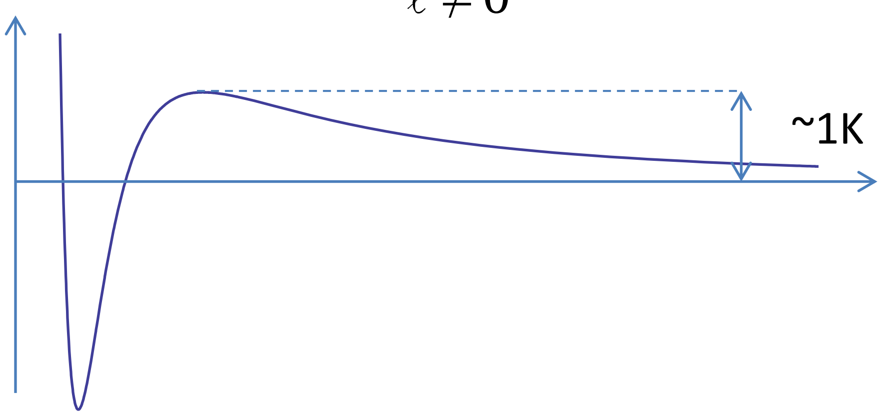
u solution of a **1D Schrödinger equation** $-\dfrac{\hbar^2}{2\mu} u'' + V_{\text{eff}}(r) u(r) = \dfrac{\hbar^2 k^2}{2\mu} u$

$$V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2}$$

$\ell = 0$



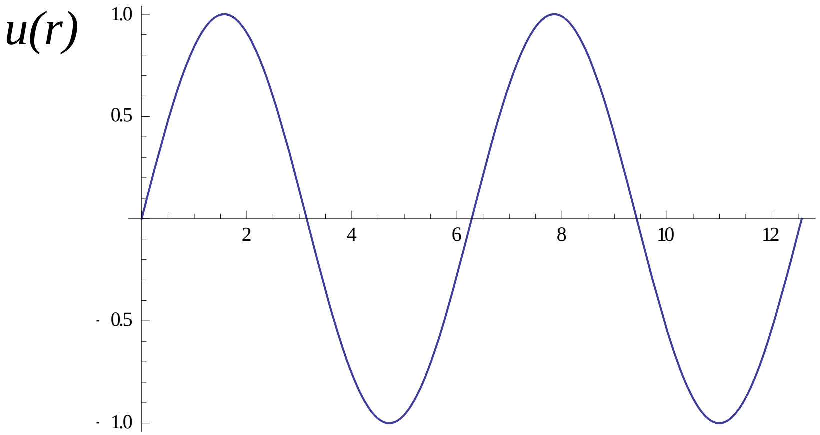
$\ell \neq 0$



At low energy, scattering occurs only in the ℓ=0 channel.

The scattering length

Free particle, ℓ=0

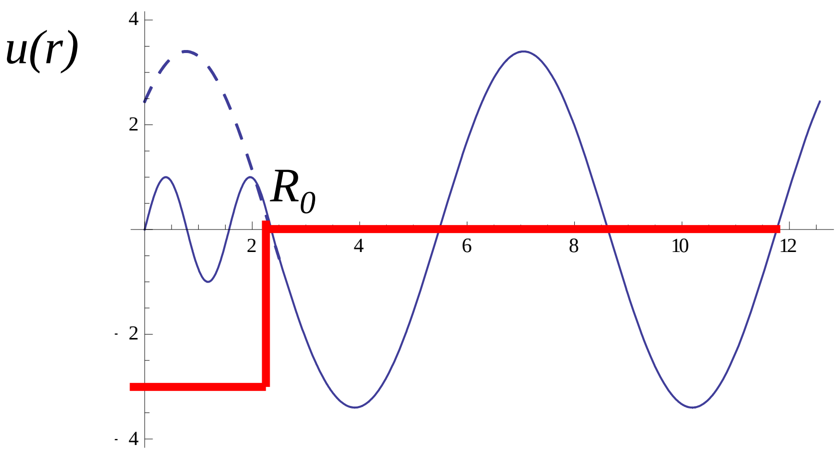


$$u(r)=\sin(kr)=\frac{e^{ikr}-e^{-ikr}}{2i}$$

Incoming

Outgoing

Short range potential



$$r>R_0 \Rightarrow u(r)=\sin(k(r-a))$$

Scattering length

Bethe-Peierls boundary condition

For a de Broglie wave-length λ>>R_0, the **matter wave does not resolve the details of the potentials** (diffraction limit).

UNIVERSALITY HYPOTHESIS : Replace in the many-body system the true potential by a simpler potential giving the same scattering length.

$$\text{Bethe-Peierls : For } \lambda_{\text{dB}} \gg r(\gg R_0),\ \psi(r) \simeq A\left(\frac{1}{r} - \frac{1}{a} + \ldots\right)$$

Alternative representations:

In 3D a Dirac-δ potential is singular and leads to UV divergencies. To cure this singularity

•Non-local pseudo-potential:

$$\langle \mathbf{r} | V_{\text{pseudo}} | \psi \rangle = g\delta(\mathbf{r})\partial_r(r\psi) \qquad g = \frac{4\pi\hbar^2 a}{m}$$

•Renormalized δ potential:

$$\langle \mathbf{r} | V_{\text{pseudo}} | \psi \rangle = g_0\delta(\mathbf{r})\psi(\mathbf{r}) \qquad \frac{1}{g_0} = \frac{1}{g} - \int \frac{d^3\mathbf{k}}{(2\pi)^3}\frac{m}{\hbar^2 k^2}$$

The scattering amplitude

$$\psi(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}} + f(k)\frac{e^{ikr}}{r}$$

$f(k)$=scattering amplitude

Scattering cross-section:

$$\sigma = 4\pi\,|\,f(k)\,|^2$$

Scattering amplitude $$f(k) = \frac{-a}{1+ika}$$

Other definition of the scattering length: $a = -\lim_{k\to 0} f(k)$

Note: even when the scattering length is infinite, the scattering amplitude remains finite.

Unitary limit: for ka>>1, the scattering amplitude remains finite $f(k) = \frac{i}{k}$

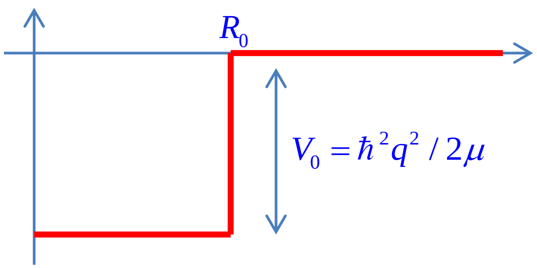
Some numbers

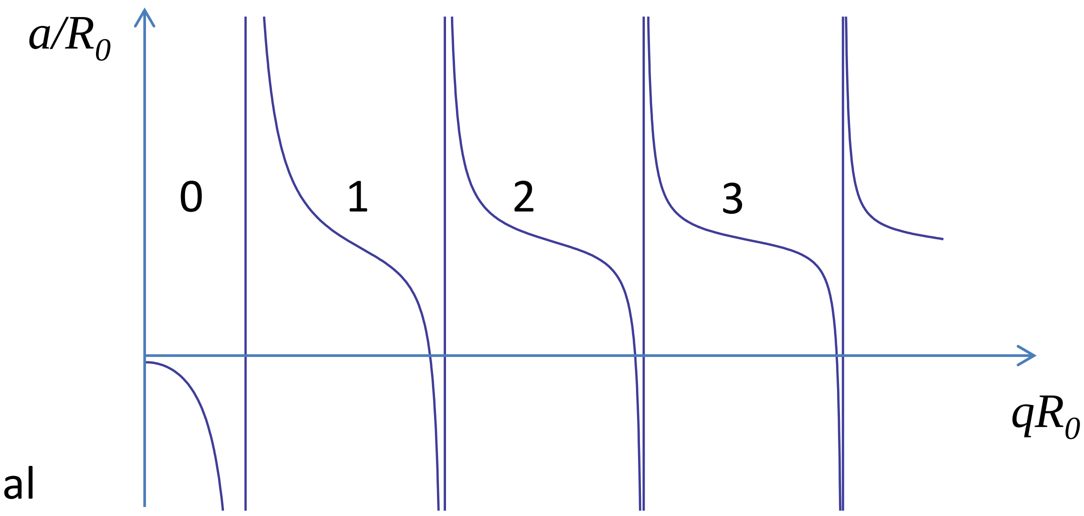
Atomic species	r_{vdw}	a (hyperfine grounstates)
^{7}Li	3.6 a_0	-27a_0
^{33}Na	4.5 a_0	52a_0
^{39}K	5 a_0	140 a_0
^{87}Rb	6 a_0	110 a_0
^{133}Cs	6.8 a_0	2400 a_0

The scattering length can differ very much from the range of the potential!

Bound states and scattering resonances

Square well potential





A scattering resonance occurs each time a bound state enters the potential

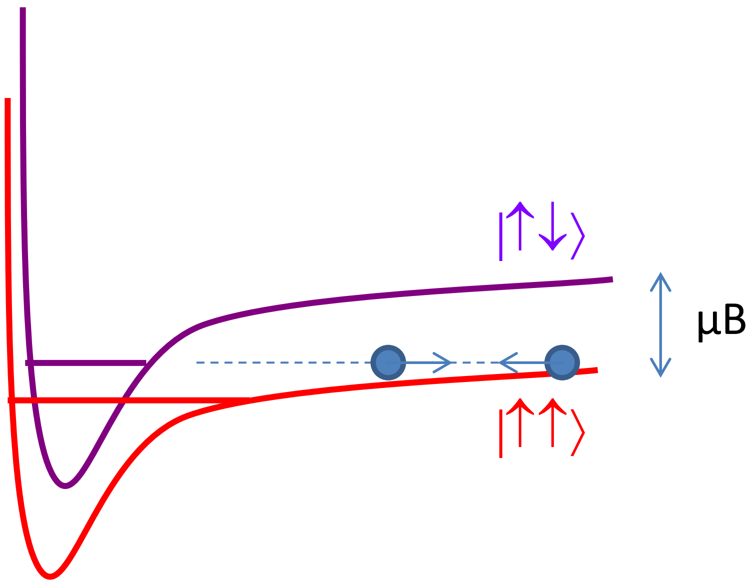
Using the Bethe-Peierls prescription:

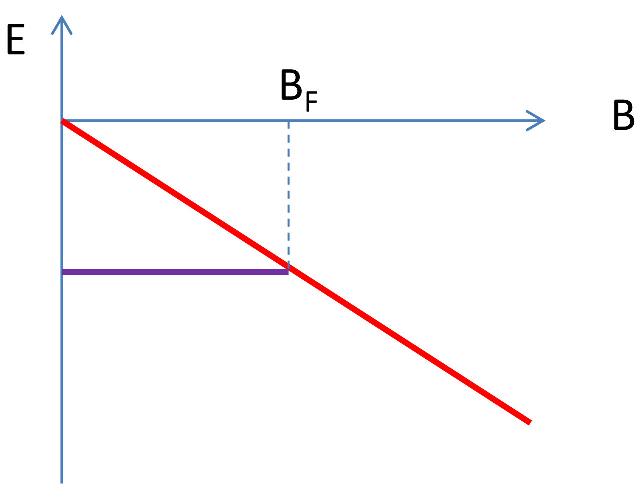
Look for bound states (E<0 solutions of Schrödinger's Equation)

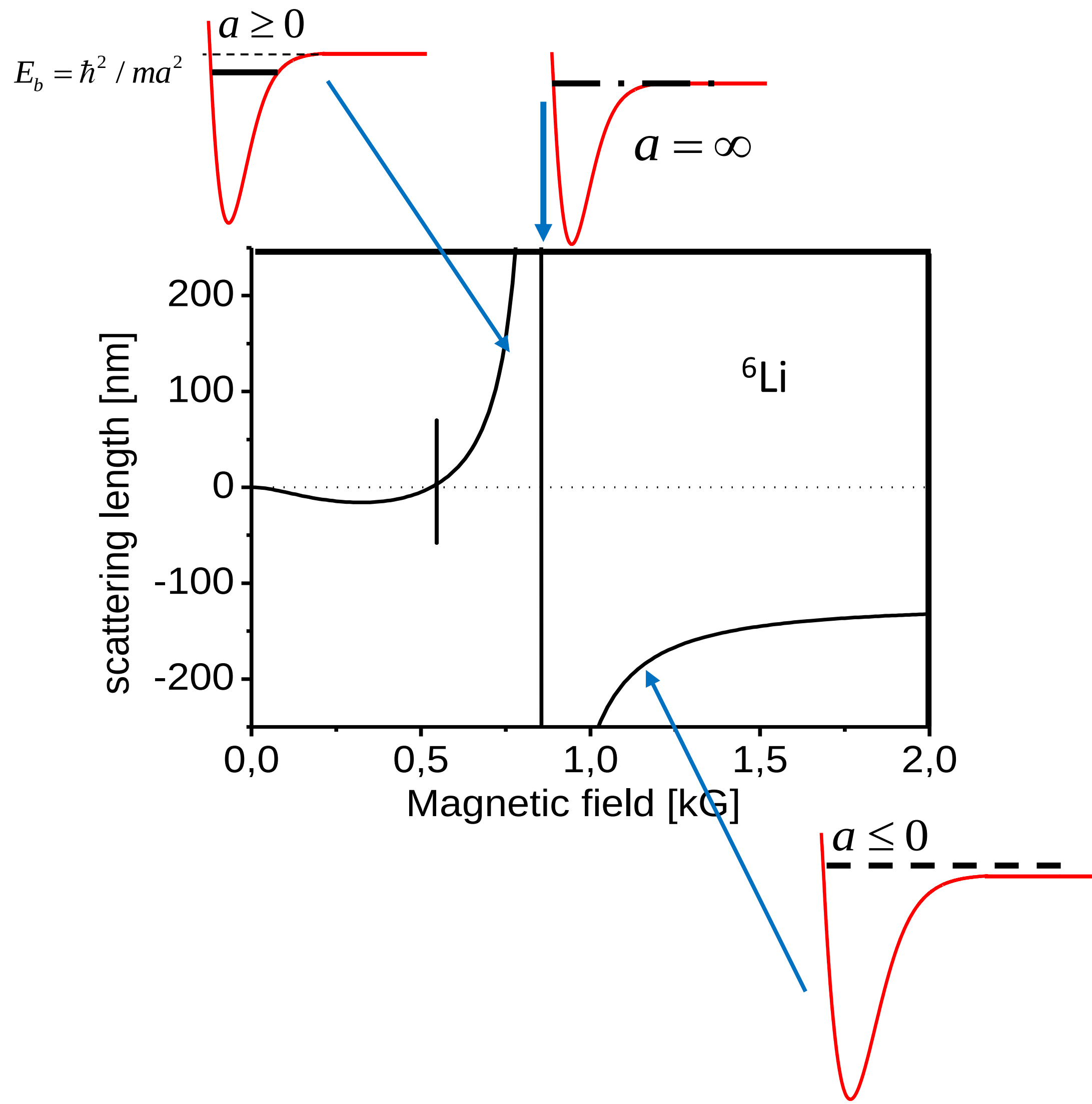
$$E = -\frac{\hbar^2 \kappa^2}{2\mu} \qquad \psi(r) = A\frac{e^{-\kappa r}}{r}$$

A bound state exists only for a>0, and then κ=a: « halo dimer »

Control of the scattering length: Feshbach resonances







Quantum statistics

Y_l^m **has the parity of** ℓ

•Fermions must scatter in odd- ℓ **wave**
•Boson must scatter in even- ℓ **wave**

At low energy, scattering in s (ℓ =0)-wave only:

no scattering between spin polarized fermions

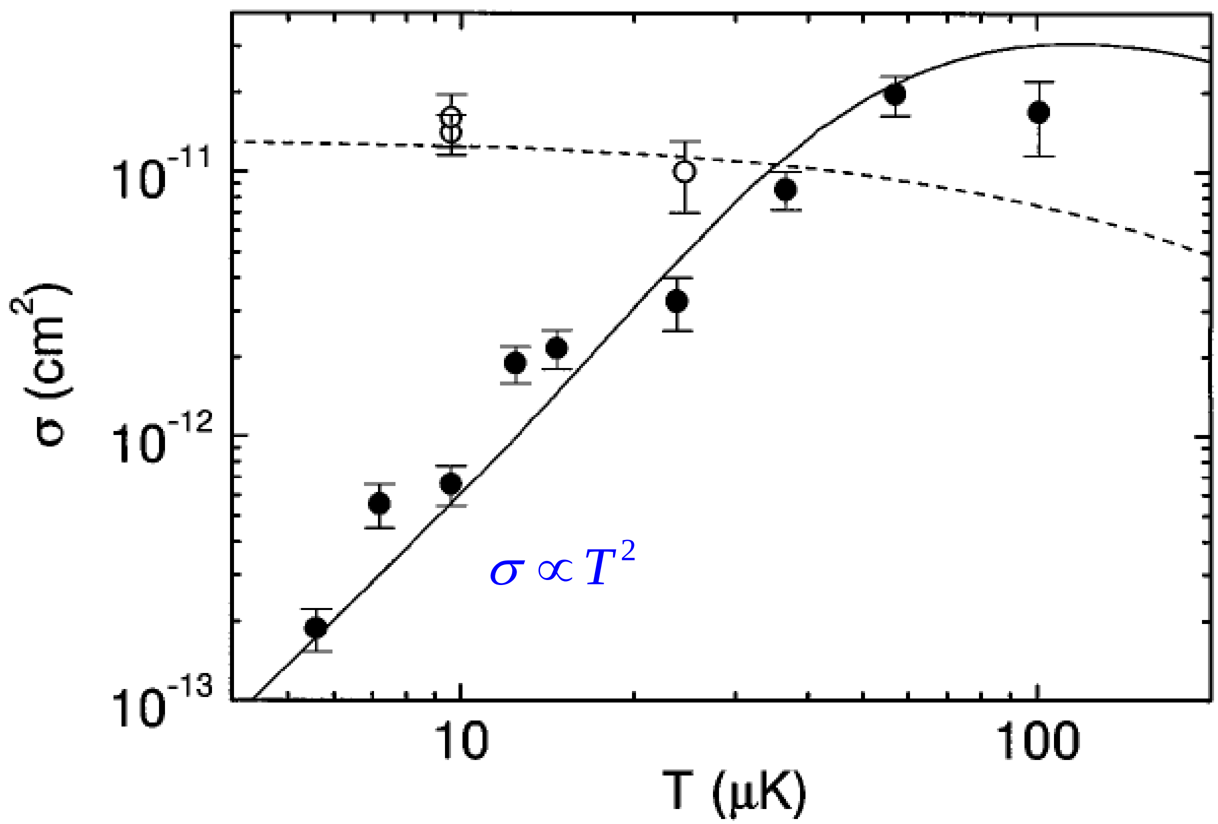
For bosons, competition kinetic energy/interaction energy.

Size of the single-particle ground state: $a_{\mathrm{ho}} = \sqrt{\frac{\hbar}{m\omega}}$

$$\frac{E_{\mathrm{int}}}{E_{\mathrm{kin}}} \simeq \frac{Na}{a_{\mathrm{ho}}} \gtrsim 10$$ for typical numbers.

Typical alkali BECs are dominated by the interaction energy.

Suppression of collisions in a spin polarized Fermi gas



B. DeMarco *et al.* Phys. Rev. Lett. **82**, 4208 (1999).